

The Multi-Armed Bandit Problem (MAB)

First, a story

You walk into a casino facing K slot machines. Each machine has its own winning probability, but you know none of them. You have T pulls (one coin each) and want to maximize your total payout.

The core dilemma: which machine do you feed your coins?

- Play the one that *looks* good — you may be misled by early luck and forever miss the truly best machine;
- Try every machine — pulling the bad ones is pure waste.

This is the **exploration vs. exploitation** trade-off, the wellspring of every problem in reinforcement learning.

Formalization

A K -armed Bernoulli bandit is defined by:

$$\text{Bandit}_K = \{\mu_1, \mu_2, \dots, \mu_K\}, \quad \mu_k \in [0, 1]$$

- μ_k — the true winning probability of arm k (**unknown**);
- action $a_t \in \{1, \dots, K\}$ — which arm to pull at step t ;
- reward $r_t \in \{0, 1\}$ — the realized payout, $r_t \sim \text{Bernoulli}(\mu_{a_t})$;
- optimal arm $\mu^* = \max_k \mu_k$, with index $k^* = \arg \max_k \mu_k$.

Objective: over T steps, choose the action sequence $(a_1, \dots, a_T)$ that maximizes expected cumulative reward $\mathbb{E} \left[\sum_{t=1}^T r_t \right]$.

Regret — the yardstick for algorithms

Definition

Maximizing reward is equivalent to minimizing **regret** — how much you lose for not knowing the best arm.

If step t pulls arm a_t , the **instantaneous regret** is:

$$\rho_t = \mu^* - \mu_{a_t}$$

and the **cumulative regret** after T steps:

$$R(T) = \sum_{t=1}^T \rho_t = T\mu^* - \sum_{t=1}^T \mu_{a_t}$$

What regret tells us

Growth rate of regret	Meaning
$R(T) = O(T)$ (linear)	the algorithm is not learning — it keeps making mistakes
$R(T) = O(\log T)$ (logarithmic)	the algorithm is learning — mistakes grow ever rarer
$R(T) = O(\sqrt{T})$ (square root)	in between the two

The Lai & Robbins lower bound (1985): any uniformly good algorithm (sublinear regret on every instance) incurs at least $\Omega(\log T)$. Note that $O(\log T)$ is only rate-optimal; matching the constant of the bound $(\sum_k \Delta_k / \text{KL}(\mu_k, \mu^*))$ is what asymptotic optimality means.

Greedy and ε -greedy

Purely greedy

The most naive idea: always pull the arm with the best current estimate.

$$a_t = \arg \max_k \hat{\mu}_k(t), \quad \hat{\mu}_k(t) = \frac{\text{total reward collected from arm } k}{\text{number of times arm } k \text{ was pulled}}$$

Fatal flaw: the first time a suboptimal arm happens to pay out $r = 1$, it is estimated at $\hat{\mu}_k = 1.0$. Pure greedy then never touches another arm again — it is stuck on a suboptimal choice forever, and regret grows linearly.

ε -greedy

Core idea: exploit greedily most of the time; explore uniformly with a small probability.

$$a_t = \begin{cases} \text{choose } k \in \{1, \dots, K\} \text{ uniformly,} & \text{with probability } \varepsilon \text{ (explore),} \\ \arg \max_k \hat{\mu}_k(t), & \text{with probability } 1 - \varepsilon \text{ (exploit).} \end{cases}$$

The incremental mean update (important!)

There is no need to store the full reward history. Given N observations with mean $\hat{\mu}_N$, a new observation r_{N+1} updates:

$$\begin{aligned} \hat{\mu}_{N+1} &= \frac{1}{N+1} \sum_{i=1}^{N+1} r_i = \frac{1}{N+1} \left(r_{N+1} + \sum_{i=1}^N r_i \right) \\ &= \frac{1}{N+1} (r_{N+1} + N\hat{\mu}_N) \\ &= \frac{1}{N+1} (r_{N+1} + (N+1)\hat{\mu}_N - \hat{\mu}_N) \\ &= \hat{\mu}_N + \frac{1}{N+1} (r_{N+1} - \hat{\mu}_N). \end{aligned}$$

$$\boxed{\hat{\mu}^{\text{new}} = \hat{\mu}^{\text{old}} + \frac{1}{N+1} (r - \hat{\mu}^{\text{old}})}$$

Why this form matters: it is the prototype of every update rule in RL:

$$\theta \leftarrow \theta + \alpha (\text{Target} - \theta)$$

Q-Learning's $Q(s, a) \leftarrow Q(s, a) + \alpha (r + \gamma \max Q - Q(s, a))$ is exactly this structure.

Regret analysis of constant ε

With fixed $\varepsilon > 0$, each step selects any specific arm with probability ε/K . A suboptimal arm k is explored in expectation $\frac{\varepsilon}{K}T$ times over T steps, each pull costing regret $\Delta_k = \mu^* - \mu_k$:

$$\mathbb{E}[R(T)] \geq \sum_{k: \Delta_k > 0} \frac{\varepsilon}{K} T \cdot \Delta_k = \frac{\varepsilon}{K} T \sum_k \Delta_k = \Omega(T)$$

Conclusion: a constant ε gives linear regret — even as $T \rightarrow \infty$, exploration never stops and keeps wasting pulls.

Decaying ε -greedy

Improvement: let the exploration probability decay over time. With $\varepsilon_t = 1/t$:

$$\mathbb{P}(\text{explore at step } t) = \frac{1}{t} \longrightarrow 0 \quad (t \rightarrow \infty)$$

Heavy exploration early (large ε_t for small t), focused exploitation late. A rigorous logarithmic-regret guarantee needs a sufficiently large decay constant: taking $\varepsilon_t = cK/(d^2t)$ (d the minimum suboptimality gap, Auer et al. 2002) yields $O(\log T)$ regret; plain $\varepsilon_t = 1/t$ only guarantees sublinear regret, and matches the rate rather than the lower-bound constant.

Caveat: the total number of explorations under $\varepsilon_t = c/t$ is about $c \ln T$; if c is too small (e.g., $c = 1$), there is barely any exploration within a finite horizon. In practice one often imposes a floor $\varepsilon_{\min}$ (such as 0.02) to retain enough exploration (see Figure 3).

Optimistic initialization

Another exploration-inducing trick: initialize all estimates at 1.0 (the maximum possible). Every arm starts overestimated — pulling an arm lowers its estimate while the others stay “inflated”, forcing the algorithm to try them all. A round of exploration happens naturally, then the algorithm converges to greedy.

Upper Confidence Bound (UCB)

The problem with ε -greedy

Its exploration is **undirected** — it picks uniformly among all arms, wasting pulls on obviously bad ones. Good exploration should be directed: “I am uncertain about this arm, so let me try it.”

Core idea: optimism in the face of uncertainty

UCB looks beyond the point estimate $\hat{\mu}_k$ and computes a **confidence upper bound**. The more uncertain an arm, the higher its bound gets pushed — even a mediocre point estimate can win selection because “it might be great”.

$$a_t = \arg \max_k [\hat{\mu}_k(t) + \text{uncertainty bonus}]$$

Hoeffding’s inequality — quantifying uncertainty

Let $X_1, \dots, X_n$ be i.i.d. random variables in $[0, 1]$ with $\bar{X}_n = \frac{1}{n} \sum_i X_i$ and $\mu = \mathbb{E}[X_i]$:

$$\mathbb{P} (|\bar{X}_n - \mu| \geq \delta) \leq 2 \exp (-2n\delta^2)$$

In plain words: after n observations, the probability that the sample mean deviates from the true expectation by $\geq \delta$ shrinks exponentially.

Run the inequality backwards: given confidence level p , solve $\exp(-2n\delta^2) = p$:

$$\delta = \sqrt{\frac{\log(1/p)}{2n}}$$

Taking $p = 1/t$ (tightening confidence as time goes on), with probability $\geq 1 - 2/t$:

$$\mu_k \in \left[\hat{\mu}_k - \sqrt{\frac{\log t}{2N_k}}, \hat{\mu}_k + \sqrt{\frac{\log t}{2N_k}} \right]$$

Select the **upper bound** (the optimistic estimate):

$$a_t = \arg \max_k \left[\underbrace{\hat{\mu}_k(t)}_{\text{exploit}} + \underbrace{c \cdot \sqrt{\frac{\log t}{2N_k(t)}}}_{\text{exploration bonus}} \right]$$

Intuition:

- a heavily pulled arm (N_k large) $\rightarrow$ small bonus $\rightarrow$ the bound tightens;

- a neglected arm (N_k small) $\rightarrow$ large bonus $\rightarrow$ the bound inflates $\rightarrow$ it gets selected again;
- $\log t$ grows so slowly with t that the long-run exploration cost stays small.

c is a hyperparameter trading off exploration and exploitation. Implementations usually smooth N_k (e.g., $N_k + 1$) to avoid division by zero and to give unpulled arms maximal uncertainty, ensuring every arm is tried at least once.

Regret upper bound (Auer et al. 2002)

UCB1 ($c = 1$) guarantees:

$$\mathbb{E}[R(T)] \leq 8 \sum_{k: \mu_k < \mu^*} \frac{\log T}{\Delta_k} + C, \quad \text{i.e., } O(\log T)$$

which is asymptotically optimal in rate. Each suboptimal arm is pulled in expectation $\approx \frac{\log T}{\Delta_k^2}$ times before its confidence interval is too narrow to compete; afterwards it is all but forgotten.

Thompson Sampling

Frequentist vs. Bayesian

The first three algorithms (Greedy, ε -Greedy, UCB) take a **frequentist** view: compute a point estimate $\hat{\mu}_k$ from history, then add something around it (noise / confidence bounds).

Thompson Sampling is **Bayesian**: instead of maintaining “one number”, maintain the **full probability distribution** of μ_k — expressing “what I believe about μ_k ”.

The Beta distribution and conjugate priors

The likelihood of Bernoulli trials (having observed s successes and f failures):

$$P(\text{data} \mid \mu) = \mu^s (1 - \mu)^f$$

The Beta density:

$$\text{Beta}(\theta \mid \alpha, \beta) \propto \theta^{\alpha-1} (1 - \theta)^{\beta-1}, \quad \theta \in [0, 1]$$

The two share the same functional form (a power of μ times a power of $(1 - \mu)$). Hence:

if the prior is Beta, then after seeing Bernoulli data the posterior is again Beta (conjugacy):

$$P(\mu \mid s \text{ successes}, f \text{ failures}) = \text{Beta}(\alpha_0 + s, \beta_0 + f)$$

With $\alpha_0 = \beta_0 = 1$, $\text{Beta}(1, 1) = \text{Uniform}(0, 1)$ — “I know nothing”.

From prior to posterior

$$\begin{aligned} P(\mu \mid r_{1:n}) &\propto P(r_{1:n} \mid \mu) \cdot P(\mu) \\ &\propto \mu^{\sum r_i} (1 - \mu)^{n - \sum r_i} \cdot \mu^{\alpha_0 - 1} (1 - \mu)^{\beta_0 - 1} \\ &= \mu^{\alpha_0 + \sum r_i - 1} (1 - \mu)^{\beta_0 + n - \sum r_i - 1} \\ &\propto \text{Beta}\left(\alpha_0 + \sum r_i, \beta_0 + n - \sum r_i\right). \end{aligned}$$

How the Beta distribution evolves with data:

Parameters	Meaning	Shape
$\alpha = 1, \beta = 1$	uninformative prior	uniform $U[0, 1]$
$\alpha = 10, \beta = 10$	9 wins, 9 losses	concentrated near 0.5, wide
$\alpha = 80, \beta = 20$	79 wins, 19 losses	concentrated near 0.8, narrow
$\alpha = 200, \beta = 50$	199 wins, 49 losses	concentrated near 0.8, narrower (very confident)

More data, narrower distribution — the belief about μ_k becomes more certain.

The experimental view of posterior convergence under Thompson Sampling is shown in Figure 1.

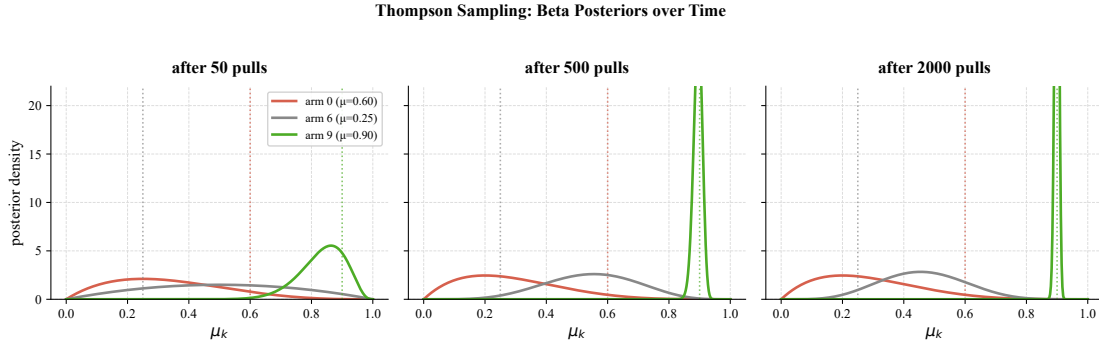


Figure 1: Posterior evolution under Thompson Sampling (10-armed Bernoulli bandit, optimal arm $\mu^* = 0.90$, fixed seed). Posteriors $\text{Beta}(\alpha_k, \beta_k)$ of three arms at three moments (steps 50 / 500 / 2000): the optimal arm (green) is pulled heavily and its posterior tightens rapidly around 0.90; a neglected suboptimal arm (gray) stays wide — posterior width is exactly what drives exploration in Thompson Sampling. Dotted lines mark the true μ_k of each arm.

The algorithm: three lines

1. **Sample:** draw one sample $\tilde{\mu}_k \sim \text{Beta}(\alpha_k, \beta_k)$ from each arm's posterior;
2. **Select:** $a_t = \arg \max_k \tilde{\mu}_k$ (greedy on the samples);
3. **Update:** $\alpha_{a_t} \leftarrow \alpha_{a_t} + r_t$, $\beta_{a_t} \leftarrow \beta_{a_t} + (1 - r_t)$.

Why does “sample from the posterior, then act greedily” work?

Probability matching: the probability of selecting arm k equals the posterior probability that arm k is the best:

$$\mathbb{P}(a_t = k) = \mathbb{P}(\mu_k = \max_j \mu_j \mid \text{data})$$

Exploration and exploitation balance automatically:

- an uncertain arm (wide Beta) $\rightarrow$ its sample swings widely, sometimes very high $\rightarrow$ it naturally gets exploration;
- a confidently bad arm (narrow Beta concentrated low) $\rightarrow$ almost never samples above a good arm $\rightarrow$ little waste.

Regret analysis (Agrawal & Goyal 2012)

Thompson Sampling also achieves expected cumulative regret $O(\log T)$; for the Bernoulli case, Kaufmann, Korda & Munos (2012) further proved that its asymptotic constant matches the Lai & Robbins lower bound. In practice it is usually the strongest performer.

Comparing the four algorithms

Experiments (Figures 2 and 3): a 10-armed Bernoulli bandit, 20 random seeds — each algorithm behaves exactly as the theory predicts.

How the exploration styles differ

- **ε -Greedy:** undirected random exploration. “The rule says explore, so I explore.”
- **UCB:** directed exploration. “I’m uncertain about arm B — what if it’s great? Give it a chance.”
- **Thompson Sampling:** probabilistic exploration. “I believe arm B has a 30% chance of being best, so I pick it 30% of the time.”
- **Decaying ε -Greedy:** same family as ε -Greedy, with a decaying exploration rate.

The fraction of pulls given to the optimal arm also exposes these differences, as shown in Figure 3.

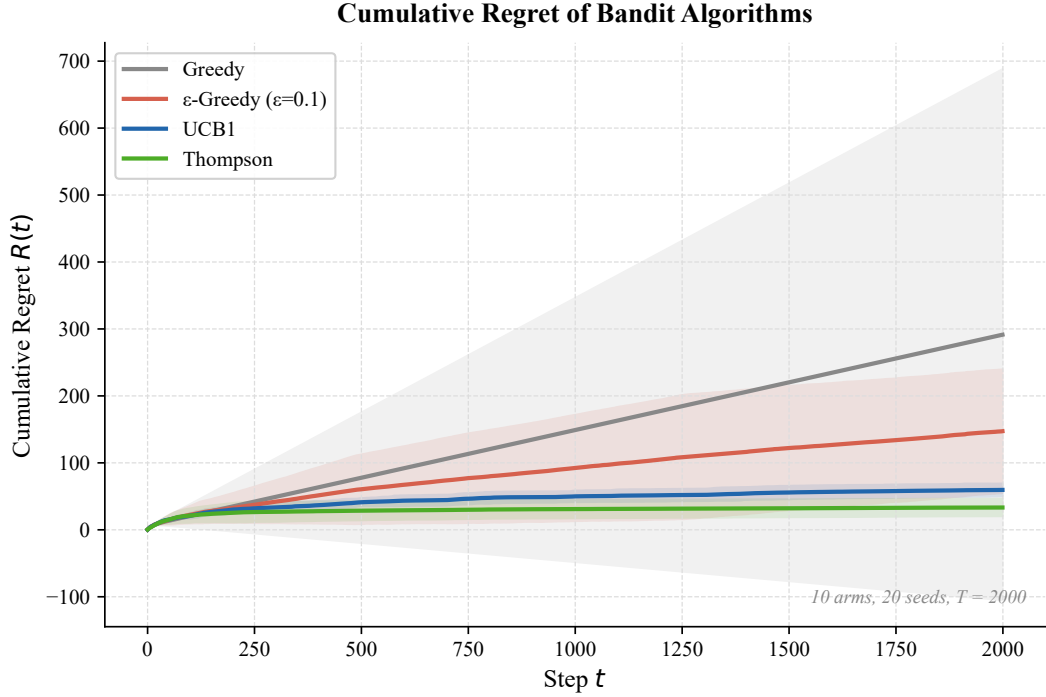


Figure 2: Cumulative regret $R(T)$ of the four algorithms (20 random seeds, shading $\pm 1\sigma$). Greedy grows nearly linearly; ε -Greedy ($\varepsilon = 0.1$) is next but still linear; UCB and Thompson Sampling are sublinear ($O(\log T)$).

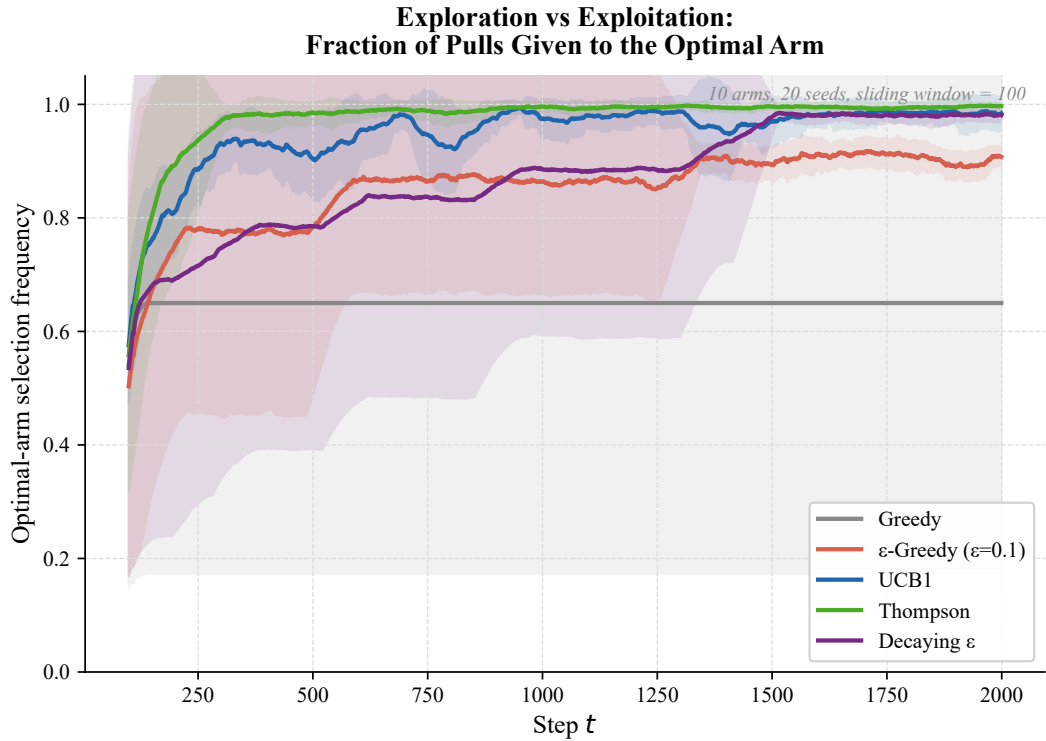


Figure 3: Fraction of pulls on the optimal arm (20 random seeds, 100-step moving average, shading $\pm 1\sigma$). Greedy locks onto a single arm per seed — here about three in ten seeds get stranded on a suboptimal arm; ε -Greedy plateaus near $1 - \varepsilon = 0.9$; decaying ε climbs slowly toward 1; UCB and Thompson Sampling converge fastest.

One table to rule them all

Algorithm	Exploration	Regret	Notes
ε -Greedy	uniformly random	$O(T)$ (linear)	simplest; regret never converges
Decaying- ε	random, decaying	$O(\log T)$ (needs tuned c)	a simple improvement
UCB	directed (bounds)	$O(\log T)$	deterministic, few knobs
Thompson Sampling	directed (sampling)	$O(\log T)$, optimal constant	strongest in practice

Practical recommendations

1. Rapid prototyping / teaching: ε -Greedy or Decaying ε -Greedy.
2. Theoretical guarantees with few knobs: UCB (a single parameter c).
3. Real applications (recommender systems, ad serving, A/B testing): **Thompson Sampling** — usually the best performer, and priors let domain knowledge flow in naturally.

Where this chapter sits in the RL landscape

The MAB is single-state reinforcement learning. Understanding it gives you the seed of three core RL concepts:

1. **Exploration vs. exploitation** — the MAB is its purest form; in an MDP it becomes “which state–action pair to explore”, an isomorphic question.
2. **Incremental updates** $\theta \leftarrow \theta + \alpha(\text{Target} - \theta)$ — the shared update skeleton of TD learning, Q-Learning, and Policy Gradient.
3. **Uncertainty-driven exploration** — UCB $\rightarrow$ count-based exploration $\rightarrow$ RND (Random Network Distillation); Thompson Sampling $\rightarrow$ PSRL (Posterior Sampling RL), the gateway to Bayesian RL.

The next chapter, MDPs, extends “single state” to “state transitions”; with the Markov property and the Bellman equations, the theoretical edifice of RL stands complete.

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